

Nishant Sharma

Email: hellonishantsh@gmail.com

New York, NY

Portfolio: hellonish.dev

LinkedIn: linkedin.com/in/nishantsh20/

Github: github.com/hellonish

Projects

Spectral Gradient Equalization (SGE) - ML Robustness Research | Spring 2026 | NYU Tandon

PyTorch · ResNet-18 · FFT/IFFT · GroupDRO · JTT · DFR · Waterbirds

- Designed SGE, a plug-in gradient intervention applying 2D FFT over convolutional kernel spatial dims to decompose weight gradients into spurious-band and causal-band components; a per-channel scaling coefficient attenuates the spurious contribution proportional to how much it dominates the gradient - no group labels required at training time
- Achieved +3.1pp worst-group accuracy over ERM and +2.5pp post-DFR WGA gain (Tail-loss + SGE) on Waterbirds benchmark (ResNet-18, 10 epochs); spurious-to-causal gradient energy ratio exceeds 1.0 from epoch 1 across all runs, confirming shortcut-frequency gradient dominance as an immediate training phenomenon
- Benchmarked 5 objectives (ERM, Group-Balanced, Tail-loss, GroupDRO, JTT) under consistent hyperparameters; diagnosed GroupDRO incompatibility as calibration stationarity failure under adversarial reweighting; designed unsupervised calibration via loss-stratified pseudo-groups to estimate the radial frequency cutoff — no ground-truth group annotations required

Singularity — AI Deep Research Platform | Mar 2026 – Apr 2026 | singularity.hellonish.dev

FastAPI · fastembed/ONNX · Qdrant · ARQ · LangGraph · PostgreSQL · Docker · AWS

- Designed 44-skill retrieval system (18 sources: ArXiv, Semantic Scholar, PubMed, SEC EDGAR, GitHub, ClinicalTrials, YouTube transcripts, PDF extraction) with fastembed/all-MiniLM-L6-v2 via ONNX — no PyTorch, no GPU at inference, 1.4GB lighter than sentence-transformers; Qdrant vector store with 0.92 cosine similarity topic cache (7-day TTL) and credibility-score payload indexing
- Implemented Phase-5 plan-before-retrieve pipeline: 3 manager agents propose full report trees in parallel → lead synthesizes → section-targeted query fanout with 2-pass source gate; StrengthConfig scales 14 parameters across 3 intensity tiers (21–130+ LLM calls); shipped solo in 13 days, ~25,500 LOC, deployed on AWS at ~\$26/month

SMOLSolver — Verifier-Guided Mathematical Reasoning | Sep 2025 – Dec 2025 | NYU Tandon

QLoRA · LoRA · PyTorch · HuggingFace PEFT · TinyLLaMA-1.1B · Phi-2 · PRM800K · GSM8K

- Owned Verifier module end-to-end: fine-tuned TinyLLaMA (1.1B) on PRM800K for step-level correctness classification (wrong/unclear/correct) via 2-phase LoRA training — Phase 1 on simple-context examples, Phase 2 on complex multi-step reasoning chains; achieved 87.91% step classification accuracy, outperforming Phi-2 backbone (81.5%) across QLoRA ablations at ranks 16 and 64
- Generator (Phi-2, 2.7B) fine-tuned on GSM8K via QLoRA (4-bit, rank=16, $\alpha=16$) achieves 64.06% Pass@1 with 7.86M trainable params (0.28%); near-perfect format compliance (99.33%); Majority@5 sampling improves accuracy +3.05%

Snap2Caption — ML Systems for Caption Generation | Mar 2025 – May 2025 | github.com/hellonish/Snap2Caption

LLaVA-1.5/1.6 · LoRA · PyTorch · MLflow · FastAPI · Prometheus · Grafana · Terraform · Ansible · Kubernetes

- Fine-tuned LLaVA-1.5-7B and LLaVA-1.6-7B (Vicuna/Mistral backends) with LoRA adapters on InstaCities1M (100k urban images, ~5GB); tracked all experiments via MLflow - BLEU, CIDEr, test loss, LoRA hyperparameters; feedback-to-retraining loop: user corrections, MinIO, Label Studio, MLflow artifact registration
- Deployed FastAPI inference server (<2s P90 latency, ~300 req/hr peak) with Prometheus/Grafana monitoring; full IaC stack: Terraform (VM provisioning), Ansible (config management), Kubernetes via Kubespray (3-node cluster)

Work Experience

Machine Learning Teaching Assistant | NYU Tandon School of Engineering | NY, USA | Sep 2025 - Present

- Standardized PyTorch training pipelines (data loading, training loops, evaluation) used by 100+ students; led debugging sessions on CNN optimization and backpropagation
- Created supplementary video lectures on ML calculus concepts (regularization, gradient updates) for graduate cohort of 50+ students

Co-Founder | Macverin Technologies | www.macverin.com | Jul 2022 - Aug 2024

- Architected distributed microservices (FastAPI + Node.js) serving 1K+ DAU for 10+ clients; maintained 99.9% system availability
- Built full-stack products end-to-end: TypeScript/React/Next.js frontends, ML inference backends, AWS deployments, automated CI/CD pipelines

Education

- New York University - Tandon School of Engineering** | New York, US | Sep 2024 - Expected May 2026
Master of Science in Computer Engineering · GPA: 3.67
FOCUS: LANGUAGE MODELS, MULTIMODAL MODELS, TRAINING & FINETUNING
- Dr. A.P.J. Abdul Kalam Technical University** | Delhi, India | Aug 2020 - Jun 2024
B.Tech Computer Science and Engineering · 1st Division with Distinction

Technical Skills

- **AI/ML:** PyTorch, LoRA/QLoRA, PEFT, SFT, RLHF (GRPO/PPO/DPO), HuggingFace Transformers, ONNX, TensorRT, MLflow, OOD robustness, spurious correlations, RAG pipelines, LangGraph, Vector stores (Qdrant, Chroma, Pinecone), fastembed, DSPy, Instructor
- **Backend:** FastAPI, PostgreSQL, Redis, ARQ, asyncpg, SQLAlchemy, WebSockets, REST, gRPC, Node.js, Docker
- **Infra:** Kubernetes, AWS (EC2, S3, RDS), Terraform, Ansible, GitHub Actions, Prometheus, Grafana
- **Frontend:** Next.js, React, TypeScript | **Languages:** Python, TypeScript, Java, C++, SQL